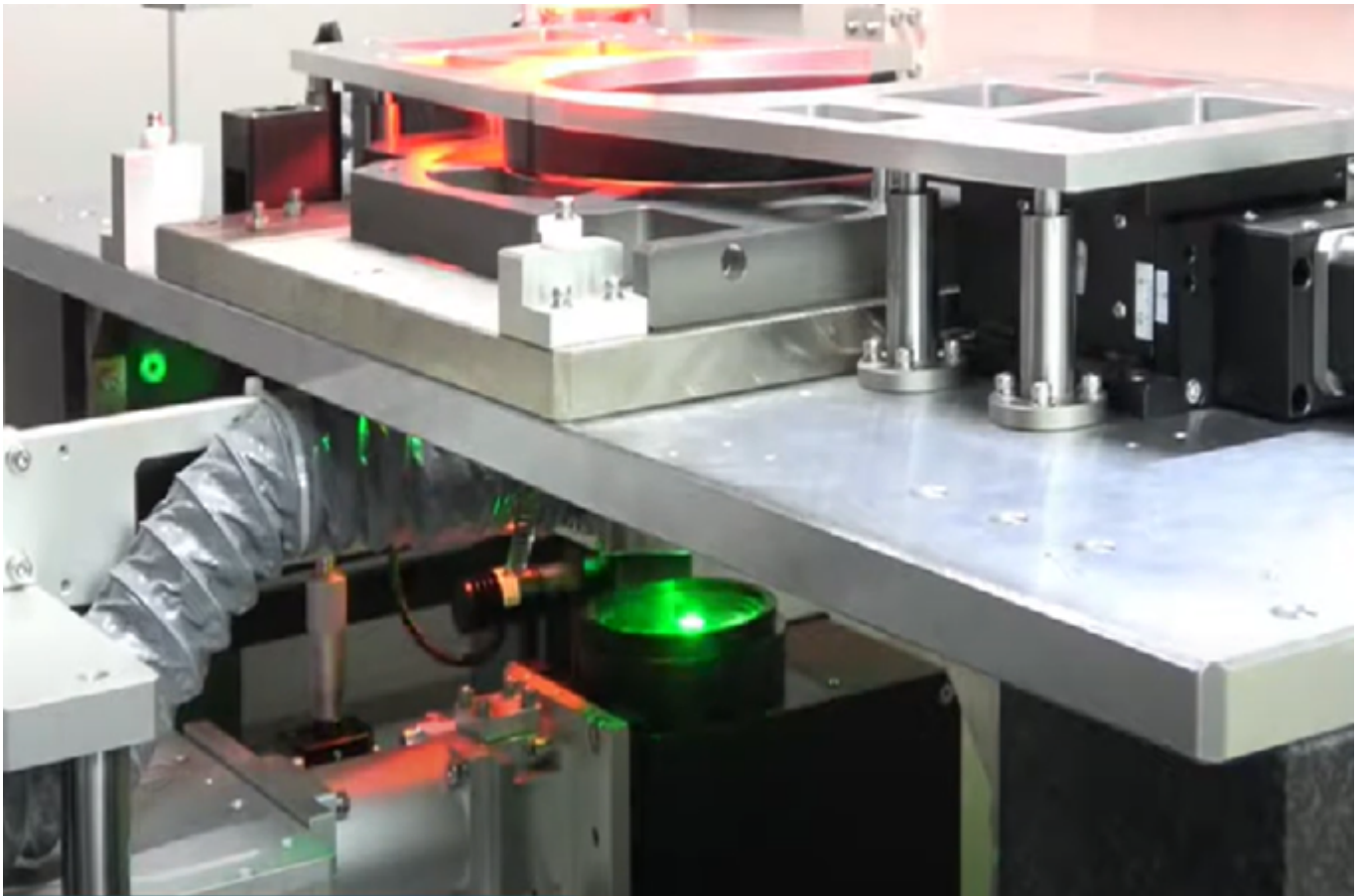


Qualification Pack



Assembly Process Sr. Technician - Laser Marking

QP Code: TEL/Q7203

Version: 1.0

NSQF Level: 5

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Qualification Pack

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Qualification Pack

TEL/Q7203: Assembly Process Sr. Technician - Laser Marking

Brief Job Description

The individual for this job role is tasked with operating laser machines to accurately etch identification marks on semiconductor wafers. The individual is also responsible to set up laser parameters, ensuring the precision of markings, and maintaining equipment. In addition, he also need to inspect marked wafers for quality and adherence to specifications, documenting the process and outcomes.

Personal Attributes

The individual must demonstrate attention to detail, analytical and problem-solving abilities, and effective communication skills, both written and verbal. The individual should be adept at working collaboratively with others.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [TEL/N7208: Operate Laser Marking Machine for Semiconductor Wafers](#)
2. [TEL/N7209: Maintain Laser Marking Equipment](#)
3. [TEL/N7210: Quality Assurance of Laser-Marked Semiconductor Wafers](#)
4. [TEL/N7211: Document Laser Marking Process](#)
5. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Telecom
Sub-Sector	Semiconductor-Manufacturing & Packaging
Occupation	Semiconductor-M&P
Country	India
NSQF Level	5
Credits	18
Aligned to NCO/ISCO/ISIC Code	NCO-2015/3522.9900

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Minimum Educational Qualification & Experience	Graduate (2nd year of 3 to 4 year UG) with NA of experience OR Completed 2nd year diploma after 12th (In relevant field) with NA of experience OR 12th Class (Electronics/Telecom/IT) with 3 Years of experience In relevant field OR Previous relevant Qualification of NSQF Level (level 4.5) with 1.5 years of experience
Minimum Level of Education for Training in School	12th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	28/02/2026
NSQC Approval Date	22/08/2024
Version	1.0
Reference code on NQR	QG-05-TL-02955-2024-V1-TSSC
NQR Version	1.0

Qualification Pack

TEL/N7208: Operate Laser Marking Machine for Semiconductor Wafers

Description

This OS unit covers operating a laser marking machine to etch identification marks on semiconductor wafers according to specifications, ensuring quality and adherence to safety protocols.

Scope

The scope covers the following :

- Pre-Operation System Checks and Preparation
- Wafer Loading, Laser Parameter Setting, and Marking Execution
- In-Process Quality Control and Minor Troubleshooting

Elements and Performance Criteria

Pre-Operation System Checks and Preparation

To be competent, the user/individual on the job must be able to:

- PC1.** power on the laser marking machine and associated equipment following SOPs
- PC2.** perform system initialization procedures as per manufacturer's instructions
- PC3.** load and align calibration target according to SOPs
- PC4.** conduct a test marking run to verify laser beam quality and system performance
- PC5.** verify proper functioning of safety interlocks and emergency stop buttons
- PC6.** review and ensure the availability of required consumables (e.g., lenses, cleaning supplies)
- PC7.** document pre-operation checks in the designated logbook

Wafer Loading, Laser Parameter Setting, and Marking Execution

To be competent, the user/individual on the job must be able to:

- PC8.** wear appropriate personal protective equipment (PPE) as per safety regulations (e.g., safety glasses, gloves)
- PC9.** prepare the designated loading area for the wafers, ensuring it's clean and free of contamination
- PC10.** carefully pick up a semiconductor wafer using appropriate handling tools (e.g., vacuum tweezers)
- PC11.** place the wafer on the designated loading stage of the laser marking machine, adhering to proper orientation and alignment as per SOPs
- PC12.** secure the wafer in place using the machine's clamping mechanism
- PC13.** access and review customer specifications for marking details (e.g., mark depth, character size)
- PC14.** identify appropriate settings for the specific material and marking requirements by referring to process reference documents (laser parameter tables)
- PC15.** input or adjust laser parameters (power, pulse width, repetition rate) on the machine control panel
- PC16.** perform a test marking on a sample wafer to verify the suitability of chosen parameters

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- PC17.** fine-tune parameters if necessary to achieve optimal marking quality within specification limits
- PC18.** document the final laser parameters used for the specific marking job
- PC19.** select the appropriate marking program on the machine control panel based on the customer specifications
- PC20.** initiate the laser marking process for the loaded wafer
- PC21.** monitor the marking process visually through the designated viewing window or using the machine's camera system (if available)
- PC22.** ensure the laser beam accurately marks the intended location and produces clear, consistent identification marks that meet customer requirements
- PC23.** release the clamping mechanism and unload the wafer using proper handling techniques, after marking completion
- PC24.** place the marked wafer in the designated unloading area or designated container for further processing

In-Process Quality Control and Minor Troubleshooting

To be competent, the user/individual on the job must be able to:

- PC25.** observe the marking process for any deviations from expected outcomes (e.g., uneven marking depth, burning, cracking)
- PC26.** monitor real-time data displayed on the machine control panel (e.g., power output, focus position)
- PC27.** identify any variations in mark quality and promptly adjust laser parameters (within allowable range) to maintain consistent and acceptable marking results
- PC28.** document any adjustments made to laser parameters and the reason for the adjustments
- PC29.** recognize common warning signs of potential equipment malfunctions (e.g., abnormal noises, error messages)
- PC30.** consult the troubleshooting guide or SOPs for recommended corrective actions for minor operational issues
- PC31.** implement basic troubleshooting steps to resolve the issue (e.g., cleaning lenses, restarting software)
- PC32.** report any unresolved issues or suspected major malfunctions to designated personnel for further action
- PC33.** document the troubleshooting attempts made and the outcome in the maintenance log

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** principles of laser light interaction with materials (e.g., ablation, photothermal effects)
- KU2.** different types of laser marking systems (e.g., pulsed lasers, continuous wave lasers)
- KU3.** factors affecting laser marking quality (e.g., laser parameters, material properties, surface cleanliness)
- KU4.** safety hazards associated with laser marking operations (e.g., eye exposure, laser emissions)
- KU5.** properties of common semiconductor materials (e.g., silicon, gallium arsenide)

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- KU6.** sensitivity of semiconductor materials to laser radiation (e.g., thermal damage)
- KU7.** handling procedures for semiconductor wafers to prevent contamination
- KU8.** functioning of the laser marking machine components (e.g., laser source, beam delivery system, scanning system)
- KU9.** operation of the machine control panel and user interface
- KU10.** interpretation of real-time data displayed on the machine panel (e.g., power output, focus position)
- KU11.** safe loading and unloading procedures for semiconductor wafers
- KU12.** importance of maintaining consistent marking quality
- KU13.** inspection techniques for identifying defects in laser markings (e.g., visual inspection, microscopic inspection)
- KU14.** interpretation of quality control charts and data
- KU15.** documentation procedures for marking process parameters and inspection results
- KU16.** importance of adhering to SOPs for safe and efficient operation
- KU17.** content of SOPs related to laser marking machine setup, operation, and maintenance
- KU18.** proper procedures for pre-operation checks, calibration, and system shutdown
- KU19.** identification of common warning signs of potential equipment malfunctions
- KU20.** use of troubleshooting guides and SOPs for resolving minor operational issues
- KU21.** safe practices for cleaning laser optics and performing basic maintenance tasks
- KU22.** importance of reporting unresolved issues to designated personnel

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work-related notes and records
- GS2.** read the relevant literature to get the latest updates about the field of work
- GS3.** listen attentively to understand the information/ instructions being shared
- GS4.** communicate politely and professionally
- GS5.** plan and prioritize tasks to ensure timely completion
- GS6.** coordinate with the co-workers to achieve the work objectives
- GS7.** evaluate all possible solutions to a problem to select the best one
- GS8.** take quick decisions to deal with workplace emergencies/ accidents

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Pre-Operation System Checks and Preparation</i>	8	12	-	3
PC1. power on the laser marking machine and associated equipment following SOPs	1	2	-	-
PC2. perform system initialization procedures as per manufacturer's instructions	1	2	-	1
PC3. load and align calibration target according to SOPs	1	2	-	1
PC4. conduct a test marking run to verify laser beam quality and system performance	2	2	-	-
PC5. verify proper functioning of safety interlocks and emergency stop buttons	1	1	-	-
PC6. review and ensure the availability of required consumables (e.g., lenses, cleaning supplies)	1	1	-	-
PC7. document pre-operation checks in the designated logbook	1	2	-	1
<i>Wafer Loading, Laser Parameter Setting, and Marking Execution</i>	20	20	-	4
PC8. wear appropriate personal protective equipment (PPE) as per safety regulations (e.g., safety glasses, gloves)	1	1	-	-
PC9. prepare the designated loading area for the wafers, ensuring it's clean and free of contamination	1	1	-	-
PC10. carefully pick up a semiconductor wafer using appropriate handling tools (e.g., vacuum tweezers)	1	1	-	-
PC11. place the wafer on the designated loading stage of the laser marking machine, adhering to proper orientation and alignment as per SOPs	2	2	-	1
PC12. secure the wafer in place using the machine's clamping mechanism	1	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. access and review customer specifications for marking details (e.g., mark depth, character size)	1	1	-	1
PC14. identify appropriate settings for the specific material and marking requirements by referring to process reference documents (laser parameter tables)	1	1	-	-
PC15. input or adjust laser parameters (power, pulse width, repetition rate) on the machine control panel	1	1	-	1
PC16. perform a test marking on a sample wafer to verify the suitability of chosen parameters	2	2	-	-
PC17. fine-tune parameters if necessary to achieve optimal marking quality within specification limits	1	1	-	-
PC18. document the final laser parameters used for the specific marking job	1	1	-	-
PC19. select the appropriate marking program on the machine control panel based on the customer specifications	2	1	-	-
PC20. initiate the laser marking process for the loaded wafer	1	1	-	1
PC21. monitor the marking process visually through the designated viewing window or using the machine's camera system (if available)	1	1	-	-
PC22. ensure the laser beam accurately marks the intended location and produces clear, consistent identification marks that meet customer requirements	1	2	-	-
PC23. release the clamping mechanism and unload the wafer using proper handling techniques, after marking completion	1	1	-	-
PC24. place the marked wafer in the designated unloading area or designated container for further processing	1	1	-	-
<i>In-Process Quality Control and Minor Troubleshooting</i>	12	18	-	3

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC25. observe the marking process for any deviations from expected outcomes (e.g., uneven marking depth, burning, cracking)	1	2	-	1
PC26. monitor real-time data displayed on the machine control panel (e.g., power output, focus position)	1	2	-	-
PC27. identify any variations in mark quality and promptly adjust laser parameters (within allowable range) to maintain consistent and acceptable marking results	1	3	-	1
PC28. document any adjustments made to laser parameters and the reason for the adjustments	1	2	-	-
PC29. recognize common warning signs of potential equipment malfunctions (e.g., abnormal noises, error messages)	2	2	-	-
PC30. consult the troubleshooting guide or SOPs for recommended corrective actions for minor operational issues	2	2	-	-
PC31. implement basic troubleshooting steps to resolve the issue (e.g., cleaning lenses, restarting software)	2	3	-	1
PC32. report any unresolved issues or suspected major malfunctions to designated personnel for further action	1	1	-	-
PC33. document the troubleshooting attempts made and the outcome in the maintenance log	1	1	-	-
NOS Total	40	50	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N7208
NOS Name	Operate Laser Marking Machine for Semiconductor Wafers
Sector	Telecom
Sub-Sector	
Occupation	Semiconductor-M&P
NSQF Level	5
Credits	5
Version	1.0
Last Reviewed Date	22/08/2024
Next Review Date	22/08/2025
NSQC Clearance Date	22/08/2024

Qualification Pack

TEL/N7209: Maintain Laser Marking Equipment

Description

This OS unit equips individuals with the skills to perform preventive maintenance, monitor performance, and respond to malfunctions of laser marking machines used for semiconductor wafers.

Scope

The scope covers the following :

- Perform Routine Maintenance on Laser Marking Machine
- Monitor Equipment Performance and Respond to Malfunctions

Elements and Performance Criteria

Perform Routine Maintenance on Laser Marking Machine

To be competent, the user/individual on the job must be able to:

- PC1.** conduct visual inspection to identify damage or leaks on the laser marking machine
- PC2.** clean the exterior surfaces of the machine using appropriate cleaning agents as per SOPs
- PC3.** verify functionality of safety interlocks and emergency stop buttons
- PC4.** ensure proper laser beam alignment and focus using a designated target according to SOPs.
- PC5.** analyze system logs for error messages or unusual operating parameters
- PC6.** perform cleaning of the laser marking machine's objective and focusing lenses (consider weekly or monthly based on usage)
- PC7.** inspect the laser marking machine's exhaust system for proper airflow and filter condition (consider weekly or monthly based on usage)
- PC8.** replenish consumables used in the marking process, such as cleaning supplies or compressed air
- PC9.** backup system data and settings as per SOPs (consider weekly or monthly based on IT policy)
- PC10.** document all completed maintenance tasks in the designated logbook, including the date and specific tasks performed

Monitor Equipment Performance and Respond to Malfunctions

To be competent, the user/individual on the job must be able to:

- PC11.** monitor the laser marking machine for abnormal noises, vibrations, or error messages during operation
- PC12.** observe the marking process to identify deviations from expected outcomes (e.g., uneven marking depth, burning, flickering laser beam)
- PC13.** recognize warning signs of potential equipment malfunctions identified in training or SOPs (e.g., unusual odor, excessive heat)

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- PC14.** • upon identifying a major equipment malfunction:
- a. cease operation immediately and secure the work area
 - b. document the observed malfunction details, including any error messages or symptoms
 - c. report the major equipment malfunction to designated personnel (supervisor, maintenance technician) as per SOPs
- PC15.** maintain a dedicated logbook or electronic record keeping system for documenting all maintenance activities
- PC16.** record the date, time, and specific tasks performed for all maintenance procedures
- PC17.** document any adjustments made during calibration or troubleshooting activities
- PC18.** report details of identified malfunctions, troubleshooting steps taken, and actions to designated personnel
- PC19.** ensure all records are completed accurately, legibly, and signed by the individual performing the maintenance

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** principles of safe operation and handling procedures for laser marking machines
- KU2.** proper cleaning techniques for the laser marking machine's exterior surfaces and optical components (lenses) according to SOPs
- KU3.** the function and importance of safety interlocks and emergency stop buttons
- KU4.** procedures for verifying and adjusting laser beam alignment and focus using a designated target
- KU5.** how to interpret system logs to identify potential errors or unusual operating parameters
- KU6.** the appropriate cleaning methods and frequency for the objective and focusing lenses (based on usage and SOPs)
- KU7.** the importance of maintaining proper airflow and filter condition in the exhaust system (based on usage and SOPs)
- KU8.** the types of consumables used in the marking process and the procedures for replenishing them
- KU9.** data backup procedures as outlined in IT policies (frequency may vary)
- KU10.** the importance of accurate and complete documentation for all maintenance tasks performed
- KU11.** the normal operating sounds, vibrations, and error messages associated with the laser marking machine
- KU12.** how to identify deviations from expected marking quality, such as uneven depth, burning, or flickering laser beam
- KU13.** warning signs of potential equipment malfunctions based on training and SOPs (e.g., unusual odour, excessive heat)
- KU14.** safe procedures for shutting down operations and securing the work area in case of a major malfunction
- KU15.** how to document equipment malfunctions, including error messages, symptoms, and observations

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- KU16.** the proper channels for reporting major equipment malfunctions to designated personnel (supervisor, technician)
- KU17.** record-keeping procedures for documenting all maintenance activities, including date, time, tasks performed, adjustments made, and troubleshooting steps taken
- KU18.** the importance of maintaining accurate, legible, and signed records of all maintenance activities

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work related records
- GS2.** read the relevant guides and literature to get the latest information about the field of work
- GS3.** communicate clearly and politely
- GS4.** listen attentively to understand the information/instructions being shared
- GS5.** plan and prioritize tasks to ensure timely completion
- GS6.** identify appropriate solutions to work related issues
- GS7.** take quick decisions in case of an emergency/accident

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform Routine Maintenance on Laser Marking Machine</i>	14	30	-	5
PC1. conduct visual inspection to identify damage or leaks on the laser marking machine	1	2	-	1
PC2. clean the exterior surfaces of the machine using appropriate cleaning agents as per SOPs	1	3	-	-
PC3. verify functionality of safety interlocks and emergency stop buttons	2	3	-	1
PC4. ensure proper laser beam alignment and focus using a designated target according to SOPs.	1	3	-	-
PC5. analyze system logs for error messages or unusual operating parameters	1	3	-	-
PC6. perform cleaning of the laser marking machine's objective and focusing lenses (consider weekly or monthly based on usage)	2	4	-	1
PC7. inspect the laser marking machine's exhaust system for proper airflow and filter condition (consider weekly or monthly based on usage)	1	3	-	1
PC8. replenish consumables used in the marking process, such as cleaning supplies or compressed air	1	3	-	-
PC9. backup system data and settings as per SOPs (consider weekly or monthly based on IT policy)	2	3	-	-
PC10. document all completed maintenance tasks in the designated logbook, including the date and specific tasks performed	2	3	-	1
<i>Monitor Equipment Performance and Respond to Malfunctions</i>	16	30	-	5
PC11. monitor the laser marking machine for abnormal noises, vibrations, or error messages during operation	2	3	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. observe the marking process to identify deviations from expected outcomes (e.g., uneven marking depth, burning, flickering laser beam)	2	3	-	-
PC13. recognize warning signs of potential equipment malfunctions identified in training or SOPs (e.g., unusual odor, excessive heat)	2	5	-	1
PC14. • upon identifying a major equipment malfunction: • a. cease operation immediately and secure the work area • b. document the observed malfunction details, including any error messages or symptoms • c. report the major equipment malfunction to designated personnel (supervisor, maintenance technician) as per SOPs	3	4	-	1
PC15. maintain a dedicated logbook or electronic record keeping system for documenting all maintenance activities	1	2	-	-
PC16. record the date, time, and specific tasks performed for all maintenance procedures	1	3	-	-
PC17. document any adjustments made during calibration or troubleshooting activities	2	3	-	-
PC18. report details of identified malfunctions, troubleshooting steps taken, and actions to designated personnel	2	4	-	1
PC19. ensure all records are completed accurately, legibly, and signed by the individual performing the maintenance	1	3	-	1
NOS Total	30	60	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	TEL/N7209
NOS Name	Maintain Laser Marking Equipment
Sector	Telecom
Sub-Sector	
Occupation	Semiconductor-M&P
NSQF Level	5
Credits	5
Version	1.0
Last Reviewed Date	22/08/2024
Next Review Date	22/08/2025
NSQC Clearance Date	22/08/2024

Qualification Pack

TEL/N7210: Quality Assurance of Laser-Marked Semiconductor Wafers

Description

This OS unit focuses on ensuring the quality of laser-marked semiconductor wafers through inspection, segregation, maintaining quality control records, and recommending process improvements.

Scope

The scope covers the following :

- Perform Quality Control Inspections of Laser-Marked Wafers
- Manage Quality Control Data and Implement Improvements

Elements and Performance Criteria

Perform Quality Control Inspections of Laser-Marked Wafers

To be competent, the user/individual on the job must be able to:

- PC1.** obtain the relevant quality control specifications for laser-marked wafers
- PC2.** prepare the inspection workstation with appropriate lighting and magnification tools (e.g., microscope)
- PC3.** select a sample of laser-marked wafers according to the sampling plan defined in SOPs
- PC4.**
 - systematically inspect each wafer for:
 - a. clarity of the marking (absence of smudging, blurring)
 - b. depth of the marking (meeting the specified depth range)
 - c. uniformity of the marking (consistent across the entire wafer)
- PC5.** compare inspection results to the quality control specifications
- PC6.** document inspection findings for each wafer sample
- PC7.** identify wafers that do not meet the quality control specifications, based on the inspection results
- PC8.** clearly mark or flag defective wafers for easy identification
- PC9.** segregate defective wafers from the acceptable ones following designated procedures to prevent them from being processed further
- PC10.** document the quantity and nature of the defects identified in each segregated wafer

Manage Quality Control Data and Implement Improvements

To be competent, the user/individual on the job must be able to:

- PC11.**
 - utilize designated forms or electronic systems to record inspection data, including:
 - a. date and time of inspection
 - b. lot ID or identification number of the wafers
 - c. sample size inspected
 - d. inspection results for each parameter (clarity, depth, uniformity)
 - e. number and type of defects identified
- PC12.** maintain quality control records according to SOPs, ensuring proper filing and archiving procedures
- PC13.** analyze inspection data to identify trends or recurring issues with marking quality

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- PC14.** prepare reports summarizing inspection findings and highlighting any deviations from the quality control specifications
- PC15.** report deviations from specifications to appropriate personnel (supervisor, quality control department) as per SOPs
- PC16.** investigate potential causes (e.g., laser power fluctuations, cleaning issues) for identified minor quality issues (within acceptable limits but approaching the threshold)
- PC17.** consult with relevant personnel (e.g., laser maintenance technician) to determine corrective actions
- PC18.** implement corrective actions (e.g., adjusting laser parameters, cleaning the marking system) following approved procedures
- PC19.** re-inspect the affected wafers to verify the effectiveness of the corrective actions
- PC20.** document the corrective actions taken and the results of the re-inspection
- PC21.** analyze historical quality control data to identify patterns or recurring issues with marking quality
- PC22.** research and investigate potential improvements to the laser marking process (e.g., optimizing laser parameters, implementing new cleaning techniques)
- PC23.** develop and document recommendations for process improvements with supporting data or justifications
- PC24.** present recommendations to relevant personnel (supervisor, process engineering team) for consideration and potential implementation

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** laser marking process and quality control principles for semiconductor wafers
- KU2.** interpretation of quality control specifications for laser-marked wafers (clarity, depth, uniformity)
- KU3.** proper operation and use of inspection equipment (lighting, magnification tools)
- KU4.** sampling techniques for wafer inspection as defined in SOPs
- KU5.** procedures for documenting inspection findings
- KU6.** techniques for segregating defective wafers to prevent further processing
- KU7.** importance of maintaining accurate and complete quality control records
- KU8.** data analysis methods to identify trends and recurring issues in quality control data
- KU9.** procedures for reporting deviations from quality control specifications
- KU10.** principles of corrective action for minor quality issues in laser marking
- KU11.** laser marking system operation and potential causes of quality variations (e.g., laser power fluctuations, cleaning)
- KU12.** procedures for implementing corrective actions while maintaining safety and following SOPs
- KU13.** methods for analyzing historical quality control data to identify areas for process improvement
- KU14.** research skills to identify potential improvements in laser marking processes
- KU15.** importance of documenting process improvement recommendations and supporting them with data

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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work related records
- GS2.** read the relevant guides and literature to get the latest information about the field of work
- GS3.** communicate clearly and politely
- GS4.** listen attentively to understand the information/instructions being shared
- GS5.** plan and prioritize tasks to ensure timely completion
- GS6.** identify appropriate solutions to work related issues
- GS7.** take quick decisions in case of an emergency/accident

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform Quality Control Inspections of Laser-Marked Wafers</i>	14	30	-	5
PC1. obtain the relevant quality control specifications for laser-marked wafers	2	3	-	1
PC2. prepare the inspection workstation with appropriate lighting and magnification tools (e.g., microscope)	1	2	-	-
PC3. select a sample of laser-marked wafers according to the sampling plan defined in SOPs	1	2	-	1
PC4. - systematically inspect each wafer for: a. clarity of the marking (absence of smudging, blurring) b. depth of the marking (meeting the specified depth range) c. uniformity of the marking (consistent across the entire wafer)	3	6	-	-
PC5. compare inspection results to the quality control specifications	1	4	-	1
PC6. document inspection findings for each wafer sample	1	2	-	-
PC7. identify wafers that do not meet the quality control specifications, based on the inspection results	2	4	-	1
PC8. clearly mark or flag defective wafers for easy identification	1	2	-	-
PC9. segregate defective wafers from the acceptable ones following designated procedures to prevent them from being processed further	1	3	-	-
PC10. document the quantity and nature of the defects identified in each segregated wafer	1	2	-	1
<i>Manage Quality Control Data and Implement Improvements</i>	16	30	-	5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. - utilize designated forms or electronic systems to record inspection data, including: a. date and time of inspection b. lot ID or identification number of the wafers c. sample size inspected d. inspection results for each parameter (clarity, depth, uniformity) e. number and type of defects identified	2	3	-	1
PC12. maintain quality control records according to SOPs, ensuring proper filing and archiving procedures	1	2	-	-
PC13. analyze inspection data to identify trends or recurring issues with marking quality	1	2	-	1
PC14. prepare reports summarizing inspection findings and highlighting any deviations from the quality control specifications	1	2	-	-
PC15. report deviations from specifications to appropriate personnel (supervisor, quality control department) as per SOPs	1	2	-	-
PC16. investigate potential causes (e.g., laser power fluctuations, cleaning issues) for identified minor quality issues (within acceptable limits but approaching the threshold)	2	3	-	1
PC17. consult with relevant personnel (e.g., laser maintenance technician) to determine corrective actions	1	2	-	-
PC18. implement corrective actions (e.g., adjusting laser parameters, cleaning the marking system) following approved procedures	1	2	-	1
PC19. re-inspect the affected wafers to verify the effectiveness of the corrective actions	1	2	-	-
PC20. document the corrective actions taken and the results of the re-inspection	1	2	-	-
PC21. analyze historical quality control data to identify patterns or recurring issues with marking quality	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. research and investigate potential improvements to the laser marking process (e.g., optimizing laser parameters, implementing new cleaning techniques)	1	2	-	1
PC23. develop and document recommendations for process improvements with supporting data or justifications	1	2	-	-
PC24. present recommendations to relevant personnel (supervisor, process engineering team) for consideration and potential implementation	1	2	-	-
NOS Total	30	60	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N7210
NOS Name	Quality Assurance of Laser-Marked Semiconductor Wafers
Sector	Telecom
Sub-Sector	
Occupation	Semiconductor-M&P
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	22/08/2024
Next Review Date	22/08/2025
NSQC Clearance Date	22/08/2024

Qualification Pack

TEL/N7211: Document Laser Marking Process

Description

This OS unit focuses on meticulous record-keeping of job parameters, inspection findings, and machine performance to ensure consistent quality control and facilitate process improvement for laser-marked semiconductor wafers.

Scope

The scope covers the following :

- Capture Job Details & Inspection Results
- Track Machine Performance & Maintain Records

Elements and Performance Criteria

Capture Job Details & Inspection Results

To be competent, the user/individual on the job must be able to:

- PC1.** record laser marking parameters (pulse width, repetition rate, power level, scan speed) for each job
- PC2.** document any changes made to settings during the marking process
- PC3.** utilize designated forms or electronic systems to store laser marking parameter data
- PC4.** record observations on clarity, depth, and uniformity of each inspected wafer
- PC5.** detect and record the nature and location of defects for rejected wafers
- PC6.** use designated forms or electronic systems to document inspection data
- PC7.** create reports for rejected wafers, specifying the defect and corrective actions taken (if applicable)

Track Machine Performance & Maintain Records

To be competent, the user/individual on the job must be able to:

- PC8.** track uptime, cycle time, and throughput of the laser marking machine
- PC9.** record the start/end times and reasons for machine stoppages
- PC10.** identify trends or issues affecting the machine's performance
- PC11.** create reports summarizing machine performance, downtime events, and potential areas for improvement
- PC12.** adhere to established procedures for maintaining logbooks related to the laser marking process
- PC13.** document date, time, operator, job details, and any observations made during the laser marking process
- PC14.** use standardized abbreviations and terminology as defined in the SOPs for clear and concise logbook entries
- PC15.** regularly review and archive logbooks according to established procedures

Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** principles of laser marking technology relevant to semiconductor wafer production
- KU2.** impact of laser marking parameters (pulse width, repetition rate, power level, scan speed) on marking quality
- KU3.** standard operating procedures (SOPs) for laser marking and wafer inspection
- KU4.** quality control specifications for laser-marked wafers (clarity, depth, uniformity)
- KU5.** techniques for identifying non-compliant wafers based on inspection criteria
- KU6.** designated forms or electronic systems used for capturing laser marking parameters and inspection data
- KU7.** importance of clear, concise, and accurate record-keeping in the laser marking process
- KU8.** methods for generating reports summarizing machine performance and downtime events
- KU9.** established procedures for maintaining logbooks related to the laser marking process
- KU10.** importance of using standardized abbreviations and terminology in logbook entries
- KU11.** safe operation procedures for the laser marking machine

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work related records
- GS2.** read the relevant guides and literature to get the latest information about the field of work
- GS3.** communicate clearly and politely
- GS4.** listen attentively to understand the information/instructions being shared
- GS5.** plan and prioritize tasks to ensure timely completion
- GS6.** identify appropriate solutions to work related issues
- GS7.** take quick decisions in case of an emergency/accident

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Capture Job Details & Inspection Results</i>	14	30	-	5
PC1. record laser marking parameters (pulse width, repetition rate, power level, scan speed) for each job	2	4	-	1
PC2. document any changes made to settings during the marking process	1	3	-	-
PC3. utilize designated forms or electronic systems to store laser marking parameter data	2	4	-	1
PC4. record observations on clarity, depth, and uniformity of each inspected wafer	2	6	-	1
PC5. detect and record the nature and location of defects for rejected wafers	3	6	-	1
PC6. use designated forms or electronic systems to document inspection data	2	3	-	-
PC7. create reports for rejected wafers, specifying the defect and corrective actions taken (if applicable)	2	4	-	1
<i>Track Machine Performance & Maintain Records</i>	16	30	-	5
PC8. track uptime, cycle time, and throughput of the laser marking machine	2	4	-	1
PC9. record the start/end times and reasons for machine stoppages	2	5	-	-
PC10. identify trends or issues affecting the machine's performance	3	5	-	1
PC11. create reports summarizing machine performance, downtime events, and potential areas for improvement	1	4	-	-
PC12. adhere to established procedures for maintaining logbooks related to the laser marking process	2	3	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. document date, time, operator, job details, and any observations made during the laser marking process	2	3	-	-
PC14. use standardized abbreviations and terminology as defined in the SOPs for clear and concise logbook entries	2	3	-	1
PC15. regularly review and archive logbooks according to established procedures	2	3	-	1
NOS Total	30	60	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	TEL/N7211
NOS Name	Document Laser Marking Process
Sector	Telecom
Sub-Sector	
Occupation	Semiconductor-M&P
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	22/08/2024
Next Review Date	22/08/2025
NSQC Clearance Date	22/08/2024

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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).

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5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.

6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass 70 % aggregate for the QP.

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
TEL/N7208.Operate Laser Marking Machine for Semiconductor Wafers	40	50	-	10	100	25
TEL/N7209.Maintain Laser Marking Equipment	30	60	-	10	100	25
TEL/N7210.Quality Assurance of Laser-Marked Semiconductor Wafers	30	60	-	10	100	20
TEL/N7211.Document Laser Marking Process	30	60	-	10	100	20
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	150	260	-	40	450	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.